Alexandre Dalban

Citizenship: USA, EU | **Settled Status:** UK

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Education

CentraleSupélec, *MSc in Artificial Intelligence (M2)* — Paris, FR | Sept 2025 – Nov 2026

- Courses: AI for Finance, Optimization for ML, Big Data for AI, Machine Learning, Reinforcement Learning.

Boston College, *BS in Mathematics and Computer Science* — Boston, USA | Aug 2021 – May 2025

- Mathematics: Probability, Statistics, Learning Theory, Mathematical Logic. CS: Algorithms, Machine Learning, Computer Systems.

Experience

Quantitative Finance Intern, *LeCap Asset Management (Hedge Fund)* — London, UK | Summers 2024, 2025

- Researched and validated 100+ alpha signals for systematic equity portfolio strategy, supporting portfolio construction.
- Built production Python tooling for the live trading environment, including an automated weekly reporting system evaluating 150+ signals across 8 performance metrics, replacing manual analysis.
- Built supervised forecasting pipeline spanning classical ML through transformer architectures, training on 90 proprietary indicators and Axioma factor-residualized returns across the top 1000 North American equities; best model achieved 54% daily directional accuracy vs. ∼50% coin-flip baseline.
- Developed PPO reinforcement learning agents with reward based on rolling 5/10/20-day realized Sharpe and P&L; agents output dynamic indicator weightings producing a composite prediction signal.

Applied AI Research Intern, *Siemens AG* — Munich, GER | May 2026 – Oct 2026

- Worked on automating a manual manufacturing step: finding which surfaces of a 3D part design form each feature to be machined. Took the team's model (XGBoost) from 51% to 76% of features exactly right on real parts, with one-change, cross-validated experiments tracing the gain to cleaner labels (+14 points), more data (+11) and new features (+1).
- Built an LLM agent that recognises features with no training on the task, querying each part's exact geometry through 21 tools: 77% against 61% for the trained model on 119 unseen parts from a newer batch, at under one cent per part.

Projects & Academic Research

IMC Prosperity 4 – Rank 458/18,800+ (13th in France) [*GitHub*] — Team of 3

- Traded a full algorithmic book: market making, basket arbitrage, EMA mean reversion, and options (vol-smile fitting, Black-Scholes vs. Bachelier); authored 94 of the team's 121 commits.
- Built an Optuna-based strategy optimizer (TPE/CMA-ES) with anti-overfitting validators (Deflated Sharpe Ratio, PBO/CSCV) and a dashboard of 13 market-microstructure panels (order-flow imbalance, mark-outs by counterparty, lead-lag CCF).

Prediction-Market Trading Strategies — Polymarket [*track record*] — Independent (two-person project)

- Priced Polymarket's daily BTC/ETH binaries off Deribit's options chain: Black-76 on the forward, an SVI smile fitted per expiry by least squares in total-variance space (PCHIP fallback on thin slices), and Breeden-Litzenberger on bracketing strikes for the risk-neutral probability; a d_2-space adjustment (Girsanov funding drift, variance risk premium, risk reversal, Cornish-Fisher skewness) mapped it to real-world probabilities.
- Sized with quarter-Kelly under a per-asset delta budget. The same generation matched equivalent markets across venues using sentence embeddings and an LLM; later ones added multi-outcome and 15-minute crypto markets, and a 102-feature LightGBM expected-value gate over L2 order-book and on-chain features, walk-forward-validated on self-collected order-book snapshots.
- 13 strategies traded or researched across three generations; $260k traded over 6,213 markets for +$155 net. Wrote the postmortems up publicly rather than the flattering numbers: settlement-time leakage that had held a backtest Sharpe at 9.5 when the leak-free figure was 0.85, and why the options edge died once the daily chain's narrow strike coverage forced interpolations the size of the edge.

Technical Skills

Quantitative Finance: Market making, alpha research, signal evaluation (rank IC, Newey-West, Deflated Sharpe/PBO), factor models (Axioma residualization), market microstructure (L2 order books, order-flow imbalance, mark-outs, maker/taker mechanics), derivative pricing (Black-Scholes, Black-76, Bachelier, IV surfaces, delta hedging), Kelly sizing
Programming: Python (advanced) | **Human languages:** Native French and English
Trading Systems: asyncio/aiohttp, WebSocket market data, fail-closed order gateways, pre-trade risk pipelines, event-driven backtest engines, walk-forward validation, Postgres/SQLite
Machine Learning: PyTorch, transformers/attention, LightGBM, XGBoost, Optuna, GNNs, RL (PPO, GRPO), LLM post-training (QLoRA, DPO), HPC (SLURM, A100)
Tools: Git, Linux, Docker, SLURM, LaTeX, Claude Code (agentic workflows, custom skills per repo)
Achievements: 1st, CentraleSupélec GreenAI hackathon | 2nd, HackEurope Solana prize ($1,000, Feb 2026)